

Talha Saleem

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Education

- Sep-2022 **Quaid-i-Azam University**, Islamabad, Pakistan.
Jul-2026
 - **B.S. in Bioinformatics**
 - CGPA: 3.0/4.0

Oct-2021 **Govt. Rizvia Islamia Degree College**, Haroonabad, Pakistan.
Oct-2021
 - **Pre-Medical (F.Sc.)**
 - Marks: 92% (& 95% in Matric Exam)

Summary

AI Engineer with hands-on experience building machine learning, computer vision, LLM, and speech systems, and deploying them as end-to-end full-stack web applications. Experienced across model development, backend APIs, frontend integration, and deployment. [Seeking opportunities across applied AI/ML, Computer Vision, and Full-Stack Engineering roles.](#)

Technical Skills

- LLMs & VLMs** Qwen3-1.7B, Llama 3.2-1B, BioBERT, Hugging Face Transformers, Ollama, NER, TF-IDF/BM25, Semantic Search, Text Classification, spaCy, Vector Search
- Speech & Retrieval** Faster-whisper, SpeechBrain, Speaker Diarization, CTranslate2, FAISS, Sentence Transformers
- Bioinformatics** Biopython, Sequence Alignment, Genomic Data Analysis, NCBI/PubMed Databases
- Web Front-Backend** FastAPI, WebSockets, SQLAlchemy, PostgreSQL, PyJWT, React, TypeScript, Vite, axios.
- ML/MLOps & AIOps** PyTorch, scikit-learn, NumPy, pandas, MLflow, DeepSpeed, Docker, Kubernetes, Nginx, Git, GitHub Actions, CI/CD, Prometheus, Grafana.
- Lang & Editors** Python, R, JavaScript, TypeScript, Cursor, VS Code, Jupyter Notebook

Work Experience

- Jan 2026 **AI Engineer**, *Upendi.ai*, South Korea, Remote.
- May 2026 **Tools:** Ollama, faster-whisper, spaCy, BioBERT, Sentence Transformers, FAISS
- [Trained and deployed open-weight LLMs](#) (Qwen3-1.7B, Llama 3.2-1B) via Ollama for NLP tasks.
 - Built [speech-to-text and diarization pipelines](#) using faster-whisper and SpeechBrain.
 - [Fine-tuned BioBERT](#), built ranking pipelines (spaCy, BM25) and FAISS retrieval, F1-evaluated.
- Sep 2025 **Associate Software Engineer**, Zeppelin Labs (SMC-Pvt), Islamabad, Pakistan.
- Dec 2025 **Tools:** FastAPI, SQLAlchemy, Docker, React, TypeScript, WebSockets
- Developed [REST/WebSocket APIs](#) with JWT auth, refresh tokens, and rate-limited endpoints.
 - Modeled databases with SQLAlchemy ORM and [containerized services using Docker and Nginx](#).
 - Built React/TypeScript frontends with an AudioWorklet module, validated via Vitest and pytest.

Notable Projects

- 2026 **BriefAI: Real-Time Meeting Transcription and Multilingual Summarization Platform.**
- Tools:** FastAPI, WebSockets, faster-whisper, Ollama (Qwen3-1.7B, Llama 3.2-1B), SpeechBrain, React, JWT, voice-to-text, text-to-text, full-stacked development
- Built a [full-stack app](#) (React, FastAPI) integrating AI services for [LLM fine-tuning and inference](#).
 - Built a live streaming pipeline via [FastAPI WebSockets](#) and browser AudioWorklet capture.
 - Transcribed audio via faster-whisper and diarized speakers using [SpeechBrain embeddings](#).
 - Ran LLMs ([Qwen3-1.7B](#) and [Llama 3.2-1B](#)) via Ollama for summarization and translation tasks.

- 2026 **ResumeRanker: AI Resume Parser and Candidate Matching Platform.**
Tools: Python, spaCy, TF-IDF/BM25, Docker, PostgreSQL
 ○ Built an [NER pipeline \(spaCy\)](#) to parse PDF/DOCX resumes and extract skills, education, experience.
 ○ Designed a [TF-IDF/BM25 algorithm](#) for candidate-job relevance matching and scoring.
 ○ Auto-tagged candidates across AI/ML, data science, and full-stack profiles.
 ○ Deployed a [Dockerized FastAPI](#) backend, storing ranking scores and match logs in PostgreSQL.
- 2026 **BioSearchAI: Biomedical RAG and Semantic Literature Search Platform.**
Tools: BioBERT, Sentence Transformers, FAISS, PubMed API
 ○ Built a document pipeline for PubMed abstracts/PDFs (extraction, chunking, cleaning).
 ○ [Trained BioBERT](#) for biomedical [entity extraction](#) across genes, diseases, drugs, and proteins.
 ○ Extended entity extraction further to cover organisms and clinical terminology.
 ○ Integrated [Sentence Transformers](#) with FAISS for semantic retrieval of sentence-level evidence.

Recent Projects

- Present **Vehicle Detection using custom trained YOLOv8 Model (ML/computer vision).**
Tools: YOLOv8, TensorRT, Ultralytics, OpenCV, Pytorch, Image processing, Google Colab
 ○ Trained a YOLOv8n model (FP16) with a synthetic, auto-labeled YOLO-format dataset pipeline.
 ○ Benchmarked detection performance (mAP50, precision/recall) on a held-out test set.
- Present **Malaria Parasite Detection System (ML/computer vision).**
Tools: YOLOv11, SAHI, Docker, FastAPI, Streamlit
 ○ Built a Poisson-blended synthetic pipeline and fine-tuned YOLOv11 for parasite localization.
 ○ Integrated SAHI for tiled fast inference and deployed via Dockerized FastAPI and Streamlit.
- 2025 **Scene2026: Gamified Year-Long Habit Tracker (Full stacked web app).**
Tools: React, Firebase, Tailwind CSS, Vite
 ○ Designed a 4-level nested calendar data model with real-time Firestore sync and token auth.
 ○ Built a custom rich-text editor using contentEditable with regex-based task parsing.
- 2024 **Differential Gene Expression Analysis of SCFA Treatment.**
Tools: GEOparse, scikit-learn, SciPy, Seaborn, Pandas, Matlib
 ○ Built a GEOparse pipeline with parametric/non-parametric testing and Cohen's D correction.
 ○ Clustered expression patterns using PCA and K-Means, visualized via volcano plots.
- 2023 **Protein 3D Structure Visualization.**
Tools: Biopython, RCSB PDB API, py3Dmol, Django
 ○ Built a tool to fetch/parse PDB structures via Biopython, calculating DSSP-based features.
 ○ Rendered interactive 3D molecular visualizations using py3Dmol.

Honors and Achievements

- 2025-present **Campus President**, AICP, [Artificial Intelligence Community of Pakistan](#).
 2025 **Co-Lead**, MSLA, [Microsoft](#) Student Learning Ambassadors.
 2024 **Campus Ambassador**, [Notion](#).
 2023 **Campus Ambassador**, GDSC, [Google Developer](#) Student Clubs.
 2021 **PEEF Scholarship**, HSSC, Punjab Educational Endowment Fund.
 2019 **PEEF Scholarship**, SSC, Punjab Educational Endowment Fund.

Language Proficiency and Relevant Courseworks

- Languages** Fluent in [English](#) (Professional Working Proficiency) | [Urdu](#) (Native)
Coursework Object-Oriented Programming, Data Structures & Algorithms, DBMS
Coursework Machine Learning, Bioinformatics Computing, Systems Biology, Computational Genomics

Collaborators & Mentors

Prof. Akmal Khan
 Professor in AI, IUB,
 Bahawalpur, Punjab, Pakistan
 Advisor/Mentor
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Khawar Islam
 Scientist at Neubility AI,
 University of Melbourne
 Melbourne, Australia
 Project Advisor
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M Ali Hamza
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 Seoul, Korea
 ML/AI Projects Advisor
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